

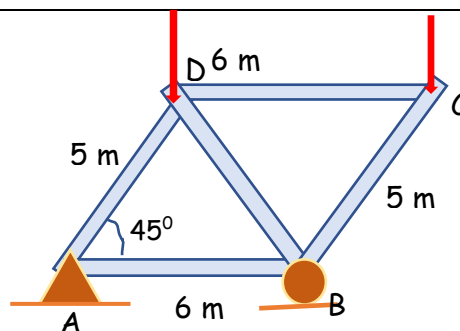
ME 19 203 ENGINEERING MECHANICS

Time 2 hrs

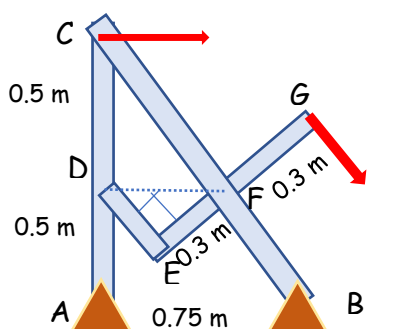
Marks 50

*** Q 1**

Find the forces in the bars of the truss if the force at C = (your roll no + 10) kN and the force at D = (your roll no + 20) kN (10 marks)

*** Q 2**

Find the forces in the joints at A, B, C, D, E, and F of the frame if the force at C = (your roll no + 10) kN and the force at G = (your roll no + 20) kN (10 marks)

*** Q 3**

Determine the principal moments of inertia of the section with respect to its centroidal axes.

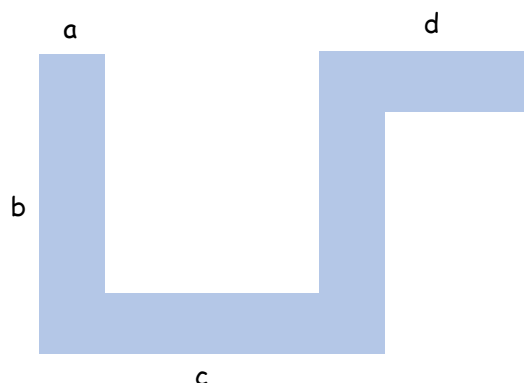
$a =$ (your roll no. + 5) mm

$b =$ (your roll no + 15) mm

$c =$ (your roll no + 20) mm

$d =$ (your roll no + 10) mm

(20 marks)

*** Q 3**

Find the moment of inertia of the shape with respect to the x and y axes (10 marks)

